Chunhui Zhang

Member of Technical Staff, Microsoft AI Superintelligence

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Experience

Microsoft AI, Superintelligence Team — Mountain View, CA

Member of Technical Staff (Researcher), RL and modelling — Jun. 2026 – Present

- Work on **RL climb for MAI-Thinking**: **scalable synthetic data generation**, **RL ablations**, and **internal evals** for agentic RL.

Amazon — Seattle, WA

Applied Scientist Intern, RL for Computer-Use (GUI) Agents — Sep. 2025 – Dec. 2025

- Post-trained multimodal LLMs (7–32B) as low-latency GUI agents with **adaptive reasoning**, using **length-control RL** to cut reasoning tokens and latency.
- Built self-distillation and a vLLM-based pipeline that backfills RL and SFT data from long, complex agent traces.

Google DeepMind — Mountain View, CA

Research Intern, Gemma 3n post-training — Jun. 2025 – Sep. 2025

- Created a synthetic audio–text data framework for real-time ASR and translation. Trained the SoTA interlaved streaming Gemma 3n.
- Built the SFT + RL training-rollout infrastructure for Gemma 3n self-distillation, making RL **5× faster**.
- Contributed to the Gemma 3n technical report (internal).

Honda Research Institute, USA — San Jose, CA

Research Intern, MLLM post-training (8B–70B) for social reasoning — Jun. 2024 – Sep. 2024

- Developed a scalable Bayesian planner for multi-step Theory-of-Mind reasoning with LLMs up to 70B+; published as an **ICML 2025 Spotlight** (top 2.59%) and filed a **US patent**.

Education

Dartmouth College, US, Ph.D. in Computer Science (advisor: Soroush Vosoughi) — Aug. 2023 – Oct. 2026

Brandeis University, US, M.S. in Computer Science; GSAS Research Fellowship — Sep. 2021 – Jun. 2023

Northeastern University, China, B.S. in Computer Science; Outstanding Honor Thesis — Sep. 2017 – Jun. 2021

Technical Skills

Post-training	RL, test time scaling, mid-training, synthetic data generation, LLM evals
Systems	PyTorch, vLLM, SGLang; distributed training and RL rollout infrastructure
Domains	Reasoning, computer-use/GUI agents

Selected Publications (* equal contribution; full list on Google Scholar)

RL, post-training, and reasoning (first author)

- [ICML 2025 Spotlight (top 2.59%)] **Overcoming Multi-step Complexity in Theory-of-Mind Reasoning: A Scalable Bayesian Planner. C. Zhang**, Z. Ouyang, S. D. Houlihan, K. Lee, N. Agarwal, S. Vosoughi, S.-Y. Lo. [paper] [code]
  Bayesian inverse planning handles global planning while the LLM does local reasoning; scales to 70B+ models.
- [ACL 2025 Oral (top 3.24%)] **Growing Through Experience: Scaling Episodic Grounding in Language Models. C. Zhang**, E. Wang, Z. Ouyang, X. Yuan, S. Vosoughi. [paper] [code]
  Post-trains 3B–70B reasoning LLMs on MCTS-sampled experience as synthentic data from physical simulators.
- [NAACL 2025 Oral (top 2.88%)] **Pretrained Image-Text Models are Secretly Video Captioners. C. Zhang***, Y. Jian*, Z. Ouyang, S. Vosoughi. [paper] [code]
  RL post-training recipe that ranked **#2** on the PapersWithCode video-captioning leaderboard, ahead of industry MLLM captioners.
- [Under review] **AdaptGUI: Adaptive Reasoning and Reward Shaping for Efficient GUI Agents. C. Zhang**, et al. [paper]
- [EMNLP 2025 Findings] **Knowing More, Acting Better: Hierarchical Representation for Embodied Decision-Making. C. Zhang**, Z. Ouyang, Z. Liu, S. Vosoughi. [paper]

- [EMNLP 2024] **Working Memory Identifies Reasoning Limits in Language Models. C. Zhang**, Y. Jian, Z. Ouyang, S. Vosoughi. [paper] [code]
- [NAACL 2025 Findings] **Temporal Working Memory: Query-Guided Segment Refinement for Enhanced Multimodal Understanding.** X. Diao*, **C. Zhang***, W. Wu, Z. Ouyang, P. Qing, M. Cheng, S. Vosoughi, J. Gui. [paper] [code]

Co-authored

- [EMNLP 2025 Oral] **SoundMind: RL-Incentivized Logic Reasoning for Audio-Language Models.** X. Diao, **C. Zhang**, et al.
- [EMNLP 2025] **Superficial Self-Improved Reasoners Benefit from Model Merging.** X. Yuan, **C. Zhang**, et al. [paper]
- [NAACL 2025 Oral (top 2.88%)] **Is It Navajo? Accurate Language Detection in Endangered Athabaskan Languages.** I. Yang, W. Ma, **C. Zhang**, S. Vosoughi.
- [ACL 2024 Oral (top 3.10%)] **Expedited Training of Visual Conditioned Language Generation via Redundancy Reduction.** Y. Jian, T. Liu, Y. Tao, **C. Zhang**, S. Vosoughi, H. Yang.
- [EMNLP 2024 Findings] **Learning Musical Representations for Music Performance Question Answering.** X. Diao, **C. Zhang**, et al.

Earlier work: robust and efficient representation learning

- [ICML 2023] **When Sparsity Meets Contrastive Models: Less Data Can Bring Better Class-Balanced Representations. C. Zhang**, et al. **AAAI-DCAA 2023 Best Paper Runner-up.**
- [ICLR 2023] **Chasing All-Round Graph Representation Robustness: Model, Training, and Optimization. C. Zhang**, et al.
- [ICLR 2024] **Aligning Relational Learning with Lipschitz Fairness.** Y. Jia*, **C. Zhang***, S. Vosoughi.
- [ICLR 2024] **Mitigating Emergent Robustness Degradation on Graphs while Scaling-up.** X. Yuan*, **C. Zhang***, et al.

Honors, Mentoring, and Service

- **Awards:** ICML Spotlight (2025); ACL Oral (2024, 2025); NAACL Oral (2025); EMNLP Oral (2025); AAAI-DCAA Best Paper Runner-up (2023); Brandeis GSAS Fellowship (2021–2023).
- **Mentoring:** mentored master's students who became co-first authors of two ICLR 2024 papers.
- **Reviewer:** ICML, ICLR, NeurIPS, ACL, NAACL, EMNLP, UAI, AISTATS, AAAI (2023–2025); TMLR, IEEE TKDE, IEEE TNSE, ACM TIST.